Tarmac Technologies

Flight Turnaround Analysis

Airline Operations Data | Feb 14–28, 2025 | 52 Turnarounds

Data Overview

Tarmac Technologies

52

Turnarounds

3

Airports
ORY · SFO · PPT

2

Aircraft Types
A350-900 · A350-1000

15

Days
Feb 14–28, 2025

39

Task Types
Tracked per flight

9

Departure-Only
Flights (no arrival)

Tarmac Technologies

Part I: Operational Performance Analysis for Airline

Business Overview

Business Insight (The Problem)	The KPI (Our Lens)
Frequent Departure Delays	**OTP Segmented by Aircraft & Airport**
Who is to Blame for Delays? ***When a flight is late, it's often unclear if ground handlers or Air Traffic Control (ATC) are responsible.***	**ADC Compliance & Delay Responsibility**
The "Domino Effect" & Hidden Costs Eg：Bag Delivery delays cause a chain reaction, leading to an estimated annual cost of €180K–€370K.	**Phase-Based Root Cause Analysis**

Data Cleaning & Preparation

1. Time Parsing
Flight times (STD/ATD/STA/ATA): converted from string "2025-02-28 - 17:40" to datetime
Task times (planning_end/actual_end): "14:43" pure time → combined with flight date
Cross-midnight fix: if task hour - STD hour > 12 → task belongs to previous day (affects 5 SFO red-eye flights)

2. Tables Built
Turnaround table: 5,564 rows → 52 rows. One row per flight with STD, ATD, ADC, ADCT
Task table: 5,564 rows → 1,186 rows. One row per turnaround × task (deduplicated from custom_label level)

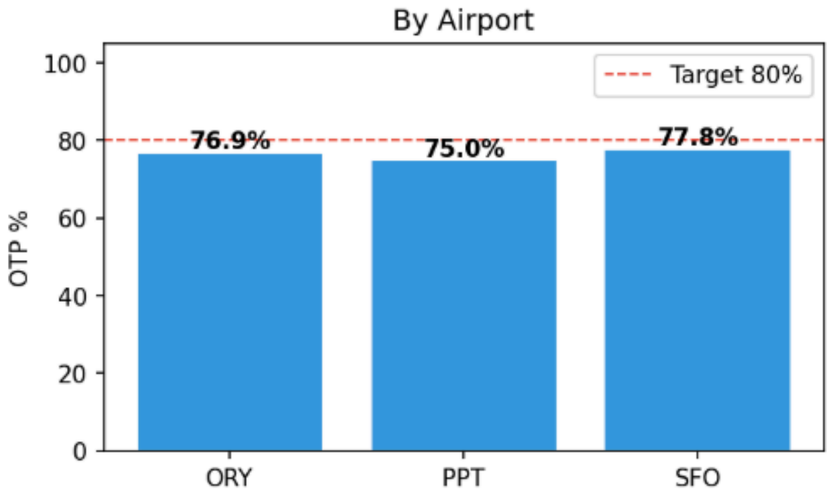
3. Null Value Policy
STA/ATA missing for 9 flights → departure-only flights, not data quality issue → kept for departure KPIs
ADC missing for 1 flight → cannot judge compliance → excluded from ADC denominator
Principle: if we can't determine the outcome, exclude from denominator — don't dilute the real rate

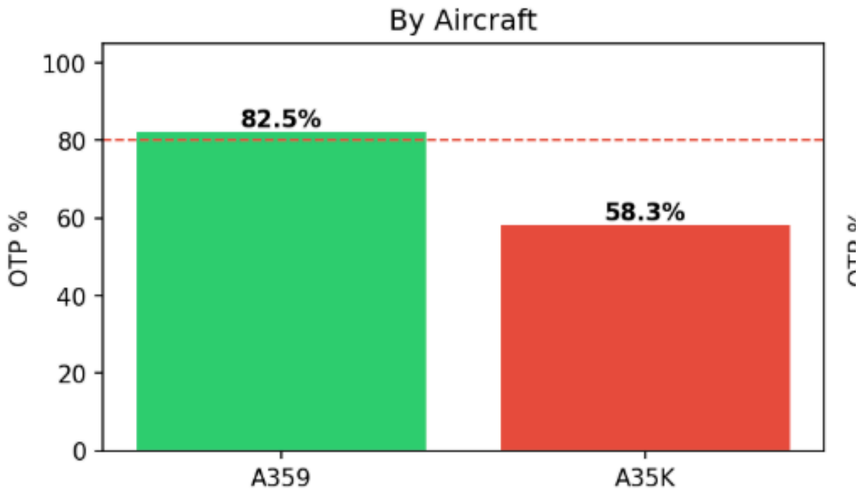
KPI 1: On-Time Performance

Tarmac Technologies

76.9%

Overall OTP
(Target: 80%)





Business Insight

- OTP 76.9% across all airports, below the 80% industry standard. No airport meets the target — this is a systemic issue, not isolated to one station.
- A35K (A350-1000) is a clear problem — 58% OTP vs 82% for A359. Larger aircraft means more baggage, cargo, and passengers to handle. The airline needs A35K-specific turnaround procedures, potentially with more ground crew or equipment. *Note: With only 12 A35K flights, this difference is not statistically significant. I recommend collecting 3+ months of data before investing in it.*

OTP — Methodology

Definition
OTP = ATD - STD ≤ 15 minutes (IATA / Eurocontrol industry standard)
ATD = Actual Time of Departure, STD = Scheduled Time of Departure

Null Handling
STD and ATD have 0 nulls → denominator = all 52 turnarounds

Why 15 minutes?
Industry convention. A flight departing 14 minutes late is considered on-time; 16 minutes is not. This threshold accounts for normal operational variance like pushback sequencing.

Dimensions
Analyzed by airport and aircraft type to dentify where and when OTP breaks down.

KPI 2: ADC Compliance

72.0%

ADC On Time
(36/50 flights)

Business Insight

- **N**early 1 in 3 flights close doors late across all airports — this is systemic, not isolated.
- A35K is worst at 54.5% vs A359's 76.9% — consistent with OTP, confirming larger aircraft turnarounds need dedicated procedures.
- 6 flights had ADC delays but still departed on time — the gap between ADCT and STD is only 3 minutes, meaning any ADC delay beyond 3 minutes directly causes an OTP miss.

ADC Compliance — Methodology

Definition

ADC Compliance = ADC ≤ ADCT (actual door close time ≤ target door close time)
Buffer = STD - ADCT

Why ADC and not just OTP?

OTP (ATD vs STD) includes pushback and ATC time after doors close — outside ground handling control.
ADC is the last moment ground ops can influence. It isolates ground handling performance from ATC factors.
NOTE:While no public industry benchmark exists for ADC (it's typically defined in ground handler SLAs).

Data Validation & Null Handling

2 flights excluded from denominator (50 instead of 52):

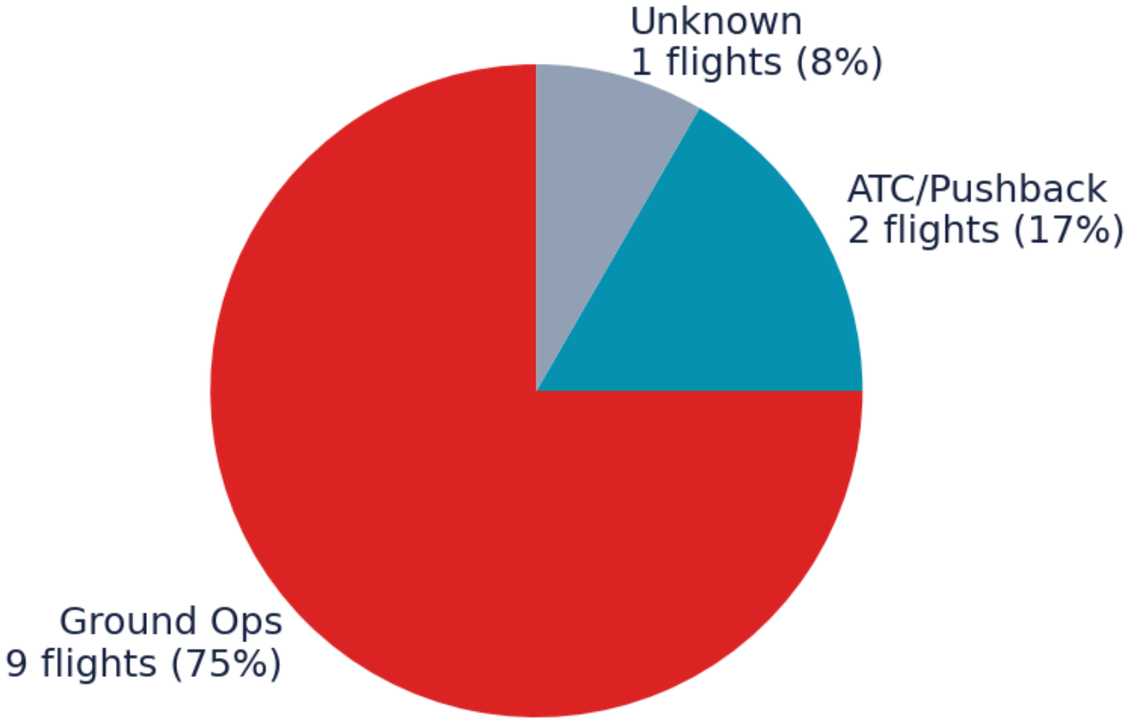
- 1 flight: ADC missing (no door-close record)
- 1 flight: ADC > ATD (doors closed after departure — logically impossible, data error)

2 flights with ADCT > STD were kept — likely updated targets after schedule changes, not data errors.

OTP vs ADC relationship

ADC and OTP are highly related but not identical.
Key difference: 6 flights were ADC-late but OTP-on-time (buffer saved them), 2 flights were ADC-on-time but OTP-late (ATC issue).

KPI 3a: Delay Responsibility



75% of delays are caused by ground operations

This means improvement efforts should focus on ground handling, not ATC coordination. The airline should review SLAs with ground service providers and investigate specific task bottlenecks.

Delay Responsibility — Methodology

Logic

For each OTP-delayed flight (ATD - STD > 15min), check ADC vs ADCT:

→ ADC > ADCT: doors closed late → ground ops caused the delay

→ ADC ≤ ADCT: doors closed on time but flight still left late → ATC/pushback issue

Why this matters

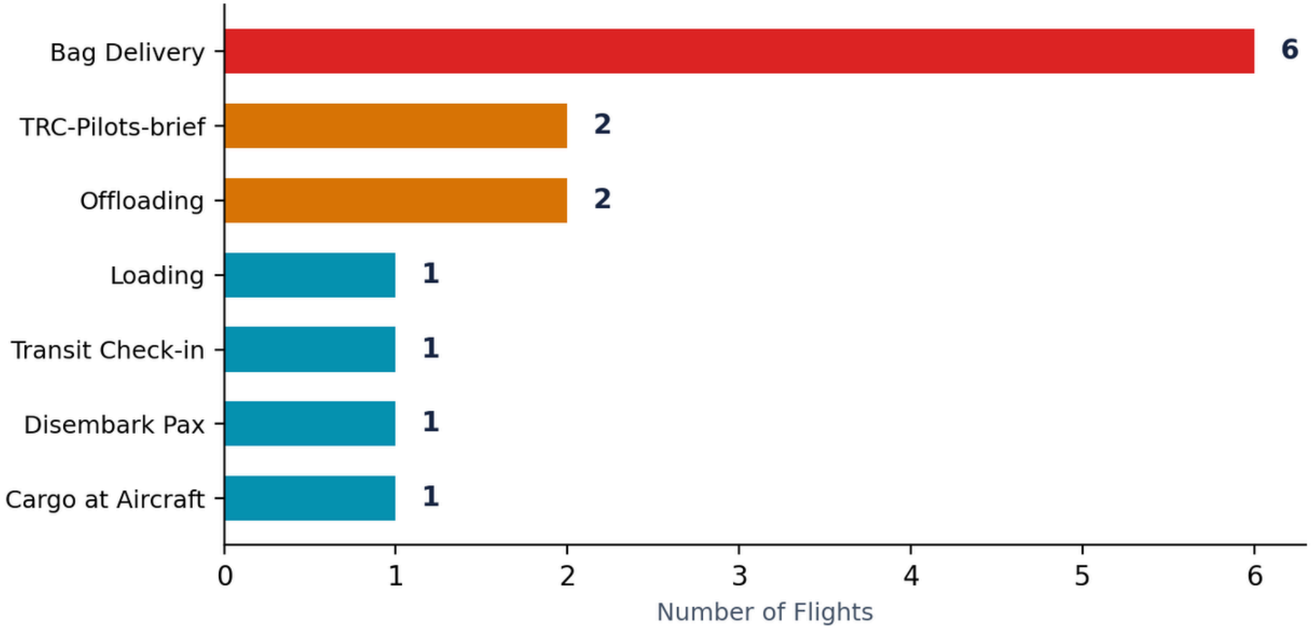
Without this split, the airline might blame ground handlers for delays actually caused by ATC slot restrictions.

Example: Flight 1625392 at SFO — ADC was 5 min early, but ATD was 62 min late. Ground ops did their job; the delay happened after door close.

Result

9 of 12 delayed flights (75%) = ground ops. Only 2 = ATC/pushback. 1 = unknown (ADC data missing).

KPI 3b: Root Cause Analysis



64%

of delays caused by baggage & cargo operations

Business Insight

- **Bag Delivery alone causes 43% of root causes (6/14). Current SLA is 24 minutes — sufficient for A359 (actual avg 17min) but not for A35K (actual avg 37min, exceeding SLA by 13min). Bag Delivery-related delays account for 33% of total delay minutes (145/444min). Annualized, this impacts ~104 flights/year. At industry-average delay cost of €50-100/min, the estimated annual impact is €180K-370K.**
- **Note: cost estimate is based on industry benchmarks, not airline-specific data.**

ORY accounts for 11 of 14 delayed flights, with Bag Delivery as the top root cause (5 times). SFO and PPT have too few delays (2 and 1) to draw airport-specific conclusions, but Bag Delivery appears in all three — reinforcing that this is a network-wide issue, not a local one.

Root Cause — Methodology

Why not just look at which task is delayed the most?

Because of cascading effects. If Bag Delivery is late → Loading is late → LDS is late → door closes late.
Blaming LDS would be wrong — it was dragged by upstream tasks.

Phase-based approach

All 39 tasks grouped into 4 phases based on operational dependency:

- Phase 1: Arrival (Disembark, Offloading, Bag Delivery...)
- Phase 2: Ground services in parallel (Cleaning, Catering, Cargo transfer...)
- Phase 3: Departure prep (FZFW, Boarding, Loading, LDS...)
- Phase 4: Door close & pushback

Root cause = first domino

For each ADC-delayed flight, scan phases in order. The first task with end_delay > 5 min = root cause.(Threshold of 5min chosen based on operational judgment. Future work: validate threshold using critical path analysis or domain expert input on typical buffers.)
All delays in later phases are treated as cascading effects, not independent causes.

Selection of delayed flights

14 ADC-delayed flights selected from 50 valid flights (2 excluded: 1 ADC missing, 1 ADC > ATD data error). We use ADC-delayed (not OTP-delayed) because tasks influence door close time, not pushback time. This also captures 'hidden' delays where buffer saved OTP.

Why use ADC-delayed flights (not OTP-delayed)?

Tasks influence door close time, not pushback time. Using ADC captures 'hidden' delays where buffer saved OTP.

Part II: Data Platform Recommendations For Tarmac

Data Integrity Findings

Tarmac Technologies

Key fields have significant null rates

STA/ATA 15.3% missing → all departure-only flights at ORY. ADC 1.8% → 1 flight with no door-close record. planning_start 60.4% → checkpoint tasks by design.

Field	Null Count	Null Rate	Reason
STA / ATA	849	15.3%	Departure-only flights
ADC	101	1.8%	Recording failure
planning_start	3,359	60.4%	Checkpoint tasks (by design)

is_punctual has a logic bug

Scheduled tasks always False (even on-time). Checkpoint tasks always True (even 149 late ones). Any dashboard metric using this field is unreliable.

Task Type	is_punctual	Rows	Issue
Scheduled	False	2,205	All False — even on-time ones
Checkpoint	True	3,359	149 actually late but marked True

task_is_applicable carries no information

All 5,564 rows are True. Either users don't use this feature, or the export filters out False values.

Value	Count	% of Total
True	5,564	**100%**
False	0	0%

Free-text fields lack standardization

Risk Identified: 9 spellings for 'no risk' (nil, NIL, N, ras, No). Bag Crew: '5/61kg', '9/125kgs', '6/84' — mixed formats. Cannot aggregate or analyze programmatically.

Field	Type	Example Values	Problem
Risk Identified	TEXT	nil, NIL, N, ras, No	9 ways to say 'no risk'
Bag Crew (Nb/poids)	TEXT	5/61kg, 9/125kgs, 6/84	Nb + weight mixed in 1 field
AVIH-D (Nb/poids)	TEXT	3/98kgs, 3/34kg, 3/21	Inconsistent units

What to do

Tarmac Technologies

Add turnaround_type to data model

9 departure-only turnarounds have no STA/ATA. Without a type tag, analysts mistake this for missing data.

Value: Enables accurate turnaround time benchmarks and prevents false data quality alerts.

Type	Count	Has STA/ATA	Arrival Tasks
Full turnaround	43	Yes	Yes
Departure-only	9	No	No

Add ADC data pipeline monitoring

1 flight has no ADC record — we only discovered this during analysis, not at ingestion time.

Value: Auto-alert on missing ADC so ops team can manually record it before the data is lost.

Current	Proposed
Missing ADC discovered during analysis	Real-time alert when ADC is not recorded within 10min of ATD
1 flight excluded from ADC compliance	Ops team notified to manually fill within same shift

Add task_type field & fix is_punctual

25 checkpoint vs 16 scheduled tasks — no field to distinguish. is_punctual broken for both.

Value: Accurate punctuality KPIs for client dashboards. Currently all metrics are wrong.

	Current	Proposed
Scheduled tasks	is_punctual = always False	True if actual_end ≤ planning_end
Checkpoint tasks	is_punctual = always True	NULL (not measurable)

Standardize free-text fields

Priority: fields we want to analyze but currently can't.

Value: Enables risk tracking, weight compliance, and cargo analytics — all currently impossible.

Field	Current	Proposed	Unlocks
Risk Identified	Free text (9 variants)	Checkbox Yes/No	Auto risk tracking
Bag Crew (Nb/poids)	'5/61kg' in 1 field	Split: NUMBER + dropdown	Weight compliance
AVIH-D (Nb/poids)	'3/98kgs' mixed	Split: NUMBER + dropdown	Cargo reporting